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Housing affordability and parental income support: The role of mortgage co-signing

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Data sources	4
2.2	Institutional environment and mortgage qualification	4
2.3	Main sample and key definitions	5
2.4	Key descriptive facts	5
2.5	Timing and outcome windows	6
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Borrower qualification constraints	6
3.2	Stylized housing-choice model without parental support	6
3.3	PTI-constrained housing and parent co-signing	7
3.4	Extension with a TDS constraint	7
3.5	Potential-outcome framework for housing choices	8
3.6	Financial-stress specification	8
3.7	Stress-test event-study specification	8
4	Identification and instruments (what makes the estimates credible)	9
4.1	What makes naive comparisons fail	9
4.2	What identifies the housing effect of co-signing	9
4.3	Why the fixed-downpayment assumption is not driving the result	9
4.4	Why the selection story is addressed credibly	10
4.5	Other credibility checks	10

4.6	Relation to the literature	10
5	Tables and figures	10
5.1	Figure 1: evolution of house prices and mortgage debt	10
5.2	Figure 2: parent co-signing relaxes the PTI constraint	11
5.3	Table 1 and Figure 3: summary statistics and the rise of co-signing	11
5.4	Figure 4, Figure 5, and Table 2: who co-signs?	12
5.5	Figure 6 and Figure 7: housing and downpayment counterfactuals	12
5.6	Figure 8 and Figure 9: leverage and payment burdens	13
5.7	Table 3, Table 4, and Figure 10: delinquency and financial stress	13
5.8	Figure 11: stress tests and causal variation in co-signing	14
6	Short summary	14
7	Conclusion	15
7.1	Core conclusion (what the paper establishes)	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** How does parental mortgage co-signing affect first-time homebuyers' ability to qualify for a mortgage, the type of housing they buy, and their subsequent financial stress?
- **Setting.** The paper studies **Canadian first-time homebuyers (FTHBs)** using linked **loan-level mortgage contract data**, **credit bureau data**, and local house-price data. The main empirical sample covers mortgages originated from **January 2015 to October 2022**, while descriptive evidence on co-signing trends goes back to **2004**.
- **Core challenge.** Parent co-signing is highly endogenous. Borrowers who co-sign with parents are different from those who do not, the data do not directly reveal how income is split across borrowers on a mortgage contract, and parents may help through gifts as well as co-signing.
- **Main contribution.** The paper makes two linked contributions:
 1. it provides the first comprehensive analysis of how **parent co-signing relaxes payment-to-income (PTI) constraints**;
 2. it identifies a new vulnerability channel: **co-signing induces adult children to lever up**, buy more expensive housing, and become more exposed to delinquency risk.
- **Why it matters.** In high-price housing markets, parental support can shift homeownership access across families. But the same support can also undermine the goal of macroprudential regulation if borrowers use the extra qualifying income to take on larger mortgages rather than safer ones.
- **Main empirical question.** Does parent co-signing simply help constrained FTHBs enter the market earlier, or does it also lead them to purchase housing beyond what they can sustainably afford on their own?
- **Main empirical results.**
 - Parent co-signing on FTHB mortgages rises from about **4% in 2004** to **13% in 2022**.
 - FTHBs co-signing with parents buy at about age **28** versus **33** for non-co-signors, and buy homes that are about **7% more expensive** on average.
 - Among parent-co-signed mortgages, about **74%** of adult children would fail to qualify for the same mortgage without parental support.
 - For these constrained co-signors, the counterfactual purchase price without parent co-signing would be about **37% lower** on average.
 - Constrained co-signors lever up heavily: the leverage statistic λ has mean about **0.76** and median about **0.89**, so chosen house prices are close to the maximum affordable with combined parent-child income.
 - Borrowers on parent-co-signed mortgages are more likely to become delinquent after origination, and this effect is strongest for **constrained** borrowers and those with high λ .
 - Quasi-experimental tightening of PTI constraints through the **2016 insured** and **2018 uninsured** mortgage stress tests increases parent co-signing, especially for borrowers close to the qualification margin.

- **Economic takeaway.** Parent co-signing relaxes a borrowing constraint, but it does not only shift timing into homeownership. It also changes the **chosen leverage point**. That can raise near-term access to housing while increasing medium-run financial fragility.

2 Data and measurement (key objects)

2.1 Data sources

- **Primary data.** The core dataset is the universe of monthly Canadian credit bureau reports from **TransUnion**. The authors observe the credit portfolios of more than **30 million Canadians**.
- **Main observed credit variables.** For each account, the data contain origination limits, balances, monthly payments, lender name, credit score, delinquency indicators (30+, 60+, 90+ days), and ownership type (sole or jointly owned/co-signed).
- **Mortgage-contract data.** The credit bureau data are matched at origination to administrative mortgage-contract data from **OSFI**, which adds contract rates, qualifying rates, qualifying income, house price, LTV, GDS, TDS, insurance status, property type, and whether the contract is a purchase, renewal, or refinancing.
- **Supplementary data.**
 - an auxiliary insured-mortgage dataset that indicates whether the FTHB received a gift for the downpayment;
 - local house-price and price-growth measures at the **forward sortation area (FSA)** level from a Teranet repeat-sales index;
 - aggregate Statistics Canada series for house prices and mortgage debt.

2.2 Institutional environment and mortgage qualification

- **Mortgage types.** Canada has **insured** mortgages (LTV above 80%) and **uninsured** mortgages (LTV at or below 80%).
- **LTV requirement.** Borrowers must have an LTV of at most **95%**. Mortgages above 80% LTV require insurance.
- **PTI constraints.** Borrowers must satisfy both a gross debt service ratio (GDS) and a total debt service ratio (TDS):

$$\text{GDS} \equiv \frac{\text{mortgage payment} + \text{property taxes} + \text{heating costs} + 50\% \text{ of condo fee}}{\text{gross qualifying income}},$$

$$\text{TDS} \equiv \frac{\text{all payments in GDS} + \text{payments on other debt obligations}}{\text{gross qualifying income}}.$$

- **Typical qualifying cutoffs.** Over the sample period, the usual maximum allowable GDS and TDS ratios are about **39%** and **44%**, respectively.
- **Credit score requirement.** Until June 2020, at least one borrower on the mortgage needed a credit score of at least **620**; afterward the minimum was raised to **680**.

2.3 Main sample and key definitions

- **Sample.** The main analysis focuses on FTHBs younger than **50** who originated purchase mortgages between **2015 and 2022**.
- **First-time homebuyer.** A borrower is classified as an FTHB if the credit bureau does not show any previous active or closed mortgage product.
- **Parent co-signing.** A mortgage is labeled parent co-signed when the age gap between the oldest and youngest borrower exceeds **18 years**. Borrowers at least 18 years older than the youngest borrower are defined as parents; the rest are adult children.
- **Income splitting assumption.** Because the contract data report only total qualifying income for the mortgage, the baseline assumption is that a parent earns the same income as an adult child, i.e. the parent-to-child income ratio (PCI) is **100%**. The paper checks robustness to a PCI of **75%** and to an age-based income profile.
- **Borrower groups.**
 - *Not parent co-signed:* **89%** of FTHBs.
 - *Parent co-signed, constrained:* **8%** of FTHBs. These are borrowers who would fail GDS/TDS/credit-score qualification without parents.
 - *Parent co-signed, unconstrained:* **3%** of FTHBs. These could qualify even without parental support under the maintained income-splitting assumption.

2.4 Key descriptive facts

- **Co-signing trend.** Parent co-signing increases steadily from about **4%** in 2004 to about **13%** in 2022.
- **Where it is strongest.** Co-signing is most prevalent in expensive cities such as **Toronto** and **Vancouver**.
- **Borrower characteristics from Table 1.**
 - Mortgages without parents have about **1.59** borrowers on average; parent co-signed mortgages have about **2.55**.
 - The youngest borrower is about **33.1** years old without parents and about **28.1** with parent co-signing.
 - Average house price is about **\$493.5k** without parent co-signing and **\$527.3k** with parent co-signing.
 - For **constrained** co-signors, average house price is even higher at about **\$582.5k**.
 - Constrained co-signors have qualifying GDS of about **27.7%** and qualifying TDS of about **41.5%**.
- **Parental balance sheets.** Co-signing parents are about **59** years old on average, often have strong credit scores, but many also carry leverage: roughly **34%** have a mortgage and **35%** have a HELOC.
- **Gifts and equity extraction.**

- In the auxiliary insured sample, more than **20%** of FTHBs receive a gift for the downpayment.
- Among co-signing parents with a mortgage or HELOC, about **33%** extracted equity within the year before co-signing.

2.5 Timing and outcome windows

- **Origination snapshot.** Qualification, income, contract terms, and co-signing status are measured at mortgage origination.
- **Delinquency horizon.** The paper studies delinquency within **two years** after origination, and also compares it to delinquency within **two years before** origination.
- **Housing-gain horizon.** Some descriptive statistics use house-price appreciation and housing gains over a **three-year** horizon after origination.

3 Models and empirical specifications (all equations + interpretation)

3.1 Borrower qualification constraints

The paper's key institutional objects are the GDS and TDS qualification rules:

$$\text{GDS} = \frac{\text{mortgage payment} + \text{property taxes} + \text{heating costs} + 50\% \text{ condo fee}}{\text{gross qualifying income}},$$

$$\text{TDS} = \frac{\text{all payments in GDS} + \text{payments on other debt obligations}}{\text{gross qualifying income}}.$$

Interpretation. Co-signing matters because it adds the parent's income to *gross qualifying income*. Mechanically, that relaxes the PTI-type underwriting constraint even if the house, price, and mortgage payment do not change.

3.2 Stylized housing-choice model without parental support

A first-time buyer with savings s and future income y chooses consumption c , housing h , and loan-to-value ratio L to maximize

$$\max_{c,h,L} \ln(c) + \sigma \ln(h - h_0)$$

subject to

$$c + RLph \leq s - (1 - L)ph + y,$$

$$(1 - L)ph \leq s.$$

The unconstrained optimum is

$$h^* = \frac{\sigma y + \sigma Rs + Rph_0}{(1 + \sigma)Rp}, \quad L^* = \frac{\sigma y - Rs + Rph_0}{\sigma y + \sigma Rs + Rph_0}.$$

The implied mortgage payment-to-income ratio is

$$\text{PTI}^* = \frac{RL^*ph^*}{y} = \frac{\sigma y - Rs + Rph_0}{(1 + \sigma)y}.$$

Interpretation. The PTI ratio becomes tighter when house prices p or borrowing costs R rise, or when the household has low income y and low savings s . This is the basic affordability mechanism in the paper.

3.3 PTI-constrained housing and parent co-signing

If lenders impose a PTI cap δ , then the maximum housing level satisfying the PTI rule is

$$\bar{h} = \frac{Rs + \delta y}{Rp}.$$

If parents co-sign and add income y_p , the PTI-feasible maximum becomes

$$\bar{h}_p = \frac{Rs + \delta y + \delta y_p}{Rp} > \bar{h}.$$

If parents also provide or promise post-origination income support equal to a fraction τy_p , then the unconstrained optimum shifts to

$$h_p^* = \frac{\sigma(y + \tau y_p) + \sigma Rs + Rph_0}{(1 + \sigma)Rp},$$

and the constrained choice with co-signing is

$$\bar{h}_p^* = \min\{h_p^*, \bar{h}_p\} > \bar{h}^*.$$

The paper defines the leverage statistic

$$\lambda = \begin{cases} \frac{\text{chosen price} - \bar{h}}{\bar{h}_p - \bar{h}}, & \text{if the FTHB co-signs and buys above } \bar{h}, \\ 0, & \text{otherwise.} \end{cases}$$

Interpretation. If $\lambda \approx 0$, co-signing is mostly a qualification device. If $\lambda \approx 1$, borrowers use nearly all of the extra parent-backed borrowing capacity to purchase a more expensive house.

3.4 Extension with a TDS constraint

The appendix extends the stylized model to include non-mortgage debt payments by children d and parents d_p . In that case,

$$\text{TDS}^* = \frac{d + \sigma y - Rs + Rph_0}{(1 + \sigma)y},$$

while with parent co-signing,

$$\text{TDS}_p = \frac{d + (1 + \sigma)d_p + \sigma y - Rs + Rph_0}{(1 + \sigma)(y + y_p)}.$$

Interpretation. Parent co-signing does not automatically relax the TDS constraint. It helps only if parents' own debt burden is not too high relative to their income. This is important empirically because many co-signing parents carry mortgages or HELOC balances of their own.

3.5 Potential-outcome framework for housing choices

Parent co-signing is treated as the treatment. The observed housing outcome for borrower i is

$$H_i = \begin{cases} H_i(0), & \text{if parents do not co-sign,} \\ H_i(1), & \text{if parents co-sign.} \end{cases}$$

The object of interest is the treatment effect on the treated,

$$ATT_i = H_i(1) - H_i(0).$$

The counterfactual $H_i(0)$ is inferred under a key assumption: **holding the downpayment fixed**, the adult child buys the most expensive house for which they would have qualified without parental support.

Interpretation. This turns regulatory underwriting rules into a measurable counterfactual. The paper does not estimate a full structural housing-demand model; instead, it asks how much housing would have been feasible without the parent’s income, debt capacity, and credit score.

3.6 Financial-stress specification

To study whether co-signing worsens ex post risk, the paper defines the change in delinquency as

$$\Delta(\text{del.}) = 100 \times [I(\text{del. within 2y post}) - I(\text{del. within 2y pre})].$$

The main regressions relate this object to a parent-co-signing indicator, interactions with whether the borrower would be constrained without parents, and interactions with λ .

Interpretation. This is designed to separate pure selection on observables from the idea that co-signing induces more leverage, and therefore more post-origination financial stress.

3.7 Stress-test event-study specification

To study whether tighter PTI constraints cause more parent co-signing, the paper estimates an event-study around the 2016 insured and 2018 uninsured stress tests:

$$y_{it} = \sum_{j=-4}^4 \gamma_j D_{it}^j + \alpha \text{Treatment}_{it} + \delta_t + X_{it}\beta + \varepsilon_{it}.$$

Here, the treatment group is composed of borrowers more likely to be close to the qualification margin, while the control group contains borrowers far from it.

Interpretation. The stress tests tighten qualifying PTI constraints for everyone, but they matter much more for high-exposure borrowers. That provides plausibly exogenous variation in the composition of who ends up co-signing with parents.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes naive comparisons fail

- Borrowers who co-sign with parents are systematically different from those who do not: they are younger, often have lower income, and often buy in more expensive neighborhoods.
- The data do not directly reveal each borrower's individual income contribution on a co-signed mortgage.
- Parents can help through gifts, equity extraction, or informal post-origination support, not only through contractual co-signing.
- Stress tests cannot be used as a simple instrument for delinquency, because the same policy directly changes the borrower's resilience and underwriting conditions.

4.2 What identifies the housing effect of co-signing

- The institutional mortgage rules are observed directly: LTV, GDS, TDS, qualifying rates, and minimum credit-score requirements are known.
- The paper simulates removing parents' **income**, **debt**, and **credit scores** from the application and recomputes whether the adult child would still qualify.
- The baseline income-splitting assumption is deliberately conservative (parent income equal to child income), and the paper shows robustness to alternative assumptions.

Identification logic. The counterfactual house price is not inferred from a reduced-form comparison group. It is inferred from the exact underwriting rules that the borrower would have faced absent parental support.

4.3 Why the fixed-downpayment assumption is not driving the result

- The paper computes the extra downpayment needed to keep the same house when parents are removed.
- Observed downpayment with co-signing averages about **\$151,204**; the counterfactual downpayment required without co-signing averages about **\$354,634**.
- In the insured sample, the median required counterfactual downpayment is about **\$147,814**, while the median gifted downpayment observed is only about **\$15,100**.
- More than **80%** of constrained insured FTHBs would need over **\$63,750** of additional downpayment to buy the same house; only about **5%** of FTHBs receive gifts that large.

Interpretation. Even allowing for gift substitution, typical parental gifts are far too small to replicate the qualifying power of co-signing for most constrained borrowers.

4.4 Why the selection story is addressed credibly

- The paper uses the **2016 insured** and **2018 uninsured** stress tests as quasi-experimental tightening of PTI rules.
- High-exposure borrowers are compared with low-exposure borrowers in an event-study/difference-in-differences design.
- Four quarters after each stress test, the share of constrained co-signers in the treatment group rises by more than **5 percentage points** relative to the control group.
- The number of parent co-signed mortgages grows faster than overall mortgage volume around the policy changes, and constrained parent co-signing grows faster still.
- Constrained co-signors before and after the stress tests do not look meaningfully different in key observables and have similar values of λ .

Interpretation. The stress tests generate additional co-signors from the margin. If those induced borrowers behave similarly to preexisting co-signors, then the paper's main mechanism is less likely to be an artifact of unobserved borrower types.

4.5 Other credibility checks

- The paper studies whether savings or longer amortization could plausibly replace co-signing. Even under optimistic assumptions, the average constrained household would need about **9.7 years** to accumulate enough extra savings, and a 40-year amortization would free only about **14.5%** of constrained borrowers from the need to co-sign.
- The paper checks alternative assumptions on how total income is split between parents and children.
- The authors also show that unconstrained co-signors are not the group driving the main stress-test results.

4.6 Relation to the literature

- The paper adds a new channel to the literature on intergenerational transfers and homeownership: parental support through **co-signing**, not just gifts or bequests.
- It connects to work on PTI/DTI constraints and macroprudential regulation by showing that tighter underwriting rules can shift borrowers toward family-backed balance-sheet arrangements.
- It also contributes to the co-signing literature by showing that for **parent** co-signing, higher leverage can dominate the insurance value of having multiple incomes on the contract.

5 Tables and figures

5.1 Figure 1: evolution of house prices and mortgage debt

Read:

- House prices in Canada rise dramatically from 2000 onward, especially in **Toronto** and **Vancouver**.
- Aggregate mortgage debt also rises sharply over the same period.
- This is the paper’s broad macro backdrop: housing affordability worsens even before the paper turns to parental support.

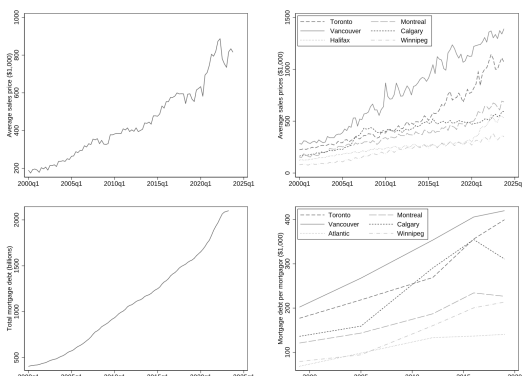


Fig. 1. Evolution of house prices and mortgage debt. Note: Fig. 1 plots the evolution of house prices and mortgage debt in Canada from 2000 to 2024. Source: Statistics Canada, including the Survey of Financial Security (1999, 2005, 2012, 2016, and 2019).

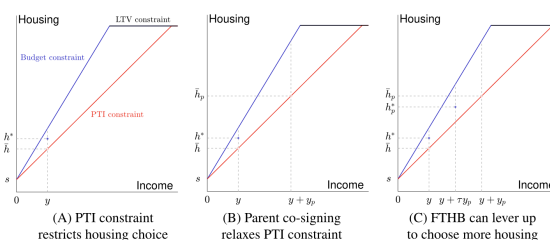


Fig. 2. Parent co-signing relaxes PTI constraint.

5.2 Figure 2: parent co-signing relaxes the PTI constraint

Read:

- Panel (A) shows that a PTI cap truncates the borrower’s feasible housing set.
- Panel (B) shows how parent income shifts out the PTI frontier.
- Panel (C) shows the paper’s key mechanism: if parents also provide ongoing support, the borrower can re-optimize and choose even more housing, not just qualify for the same house.

5.3 Table 1 and Figure 3: summary statistics and the rise of co-signing

Table 1
Summary statistics by mortgage contract types.

	Not co-signed	Parent co-signed	Co-signed: unconstrained	Co-signed: constrained
Number of borrowers	1,591	2,552	2,558	2,549
Number of parents	.	1,310	1,241	1,334
Age (oldest on contract)	34.99	58.68	57.45	59.11
Age (youngest on contract)	33.06	28.06	26.94	28.45
Credit score (parent)	.	802.2	816.4	797.0
Credit score (FTHB)	777.7	750.1	749.9	750.2
Total income (\$1000)	111.1	153.6	207.2	135.0
Loan size (\$1000)	386.5	397.2	286.4	435.8
LTV (%)	83.05	80.67	82.80	79.93
Qualifying GDS (%)	29.57	23.76	12.44	27.70
Qualifying TDS (%)	36.84	39.19	32.56	41.50
Interest rate (%)	2.594	2.614	2.641	2.605
Mortgage payment (\$)	1671.4	1702.2	1254.3	1858.0
House price (\$1000)	493.5	527.3	368.8	582.5
FSA mean house price (\$1000)	572.2	611.5	537.9	637.1
I(house price > FSA mean)	0.352	0.339	0.209	0.385
3-year FSA price growth (%)	27.49	29.35	28.29	29.74
3-year housing appreciation (\$1000)	123.0	137.9	95.46	153.8
3-year housing gain (\$1000)	96.62	110.6	74.98	123.9
3-year housing gain/FTHB income (%)	99.00	193.9	82.60	235.6
Observations	511,148	63,083	16,286	46,797

Note: Table 1 presents summary statistics for FTHB mortgages originated between January 2015 and October 2022. Column (1) presents summary statistics for mortgages without parent co-signing. The remaining columns are parent-co-signed mortgages. Column (3) shows borrowers who would likely qualify for mortgages even without parents, while column (4) shows borrowers constrained by GDS/TDS/credit score requirements without parent co-signing. Mortgage payments are monthly, while total income is annual total gross income. We calculate the housing appreciation 3 years after mortgage origination as the change in house prices given the purchase price and the FSA house price growth. Housing gain is the difference between housing appreciation and mortgage interest cost over the same period. The last row shows the ratio of 3-year housing gain to FTHB income. For each variable we report the mean value.

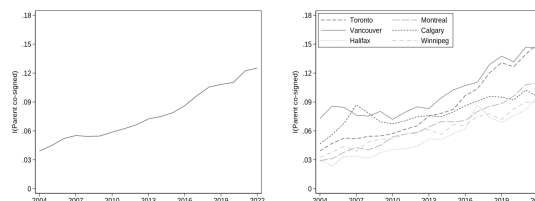


Fig. 3. Evolution of parent co-signing. Note: Fig. 3 plots the evolution of parent co-signing in Canada since 2004.

Read:

- Parent co-signing rises from about **4%** in 2004 to about **13%** in 2022.
- Mortgages with a co-signing parent have about **2.6 borrowers**, versus **1.6** without parents.
- Co-signing FTHBs are about **5 years younger** and buy more expensive homes.
- Constrained co-signors stand out as the most leveraged and most stretched subgroup inside the co-signing sample.

5.4 Figure 4, Figure 5, and Table 2: who co-signs?

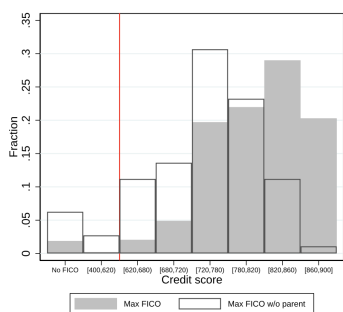


Fig. 4. Distribution of credit scores.
Note: Fig. 4 plots the distribution of credit scores for parent co-signed mortgages. The gray bars are the maximum reported credit score on the mortgage. The white bars are the maximum credit score excluding the parent score. The vertical line represents the cut-off for a mortgage to be approved.

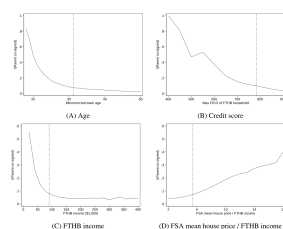


Fig. 5. Prevalence of parent co-signing across FTHB characteristics.
Note: Fig. 5 plots the prevalence of parent co-signing across FTHB age, credit, income, and the ratio of neighborhood house prices to income. We use the same variables and control variables. The vertical line represents the median of each characteristic. The age, income, and credit scores are from TransUnion while the FSA mean house price comes from Zillow.

	Dependent variable: I(parent co-sign)			
Log(FSA mean house price)	0.073*** (0.0055)	0.069*** (0.0047)	0.070*** (0.0051)	0.047*** (0.0057)
FSA price growth (%)	0.00059* (0.00024)	0.00044** (0.00014)	0.00039 (0.00022)	0.000049 (0.000094)
Log(per-borrower income)	-0.070*** (0.0070)	-0.072*** (0.0071)	-0.070*** (0.0070)	-0.072*** (0.0072)
FTHB age	-0.0094*** (0.00085)	-0.0094*** (0.00082)	-0.0094*** (0.00085)	-0.0093*** (0.00083)
Credit score (FTHB)	-0.00090*** (0.000023)	-0.00092*** (0.000021)	-0.00091*** (0.000023)	-0.00092*** (0.000021)
Year FE	N	Y	N	Y
City FE	N	N	Y	Y
R ²	0.088	0.090	0.090	0.093

Note: Table 2 reports the correlation between parent co-signing and housing prices. After dropping observations without a credit score, there are 503,915 observations. The dependent variable is an indicator that equals 1 if a mortgage is co-signed with parents, and 0 otherwise. The average house price and year-over-year house price growth at the FSA level are based on a repeat-sales index published quarterly by Zillow. Standard errors clustered at city level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Read:

- Figure 4 shows that adult children's credit scores are much lower than the maximum score on the mortgage once parent scores are excluded; around **9%** of children have missing or below-cutoff scores.
- Figure 5 shows that co-signing is more common for younger, lower-income, lower-credit-score borrowers and where local house prices are high relative to income.
- Table 2 confirms this in regressions: co-signing is positively related to local house prices and negatively related to age, income, and credit score.

5.5 Figure 6 and Figure 7: housing and downpayment counterfactuals

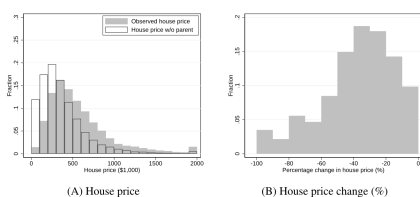


Fig. 6. Housing outcomes in the absence of parental support.
Note: Fig. 6 plots counterfactual housing outcomes for constrained FTHBs. Among parent co-signed mortgages, 74 percent of the adult children are constrained without parent support, unable to meet either the GDS/TDS requirements (72 percent) or credit score requirements (9 percent). For these constrained children, in the absence of parent support, the mean change in house price is -37 percent.

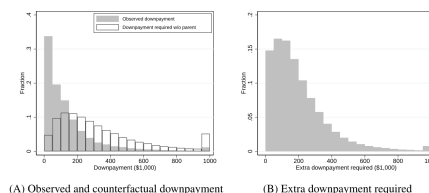
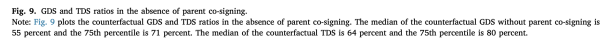
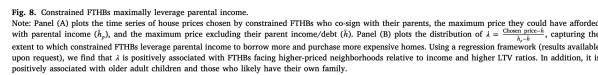


Fig. 7. Downpayment required in the absence of parent co-signing.
Note: Fig. 7 plots the counterfactual downpayment required in the absence of parent co-signing. The mean of the observed and counterfactual downpayments are \$151,204 and \$354,634, respectively. The median additional downpayment required to replace parent co-signing is \$158,235 and the mean is \$203,430.

Read:

- ## 5.6 Figure 8 and Figure 9: leverage and payment burdens



- Figure 8 shows that constrained co-signors choose houses much closer to the maximum affordable with **combined parent-child income** than to the maximum affordable with their own income alone.
- The leverage statistic λ has mean **0.76** and median **0.89**, so constrained co-signors use most of the extra borrowing capacity made available by parents.
- Figure 9 shows the implied cash-flow strain without parents: the median counterfactual GDS is about **55%**, and the median counterfactual TDS is about **64%**.

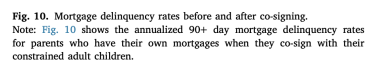
5.7 Table 3, Table 4, and Figure 10: delinquency and financial stress

Panel A: Dependent variable = Δ delinquency within 2 years after origination($\times 100$)					
	All credit products		Mortgage products		
Exposure co-signed	2.23**	1.09**	0.24**	0.13*	
1TV < 60 HS.RS	(0.14)	(0.22)	(0.03)	(0.04)	
Exposure co-signed ($\times$ contaminated)	0.21	1.59**	0.18**	0.15**	
1TV < 60 HS.RS	(0.22)	(0.30)	(0.03)	(0.05)	
<i>F</i>	0.018	0.018	0.005	0.005	
Panel B: Dependent variable = Δ delinquency within 2 years after origination($\times 100$)					
	All credit products		Mortgage products		
Exposure co-signed	1.81**	0.41	0.36**	0.22**	
1TV < 60 HS.RS	(0.30)	(0.38)	(0.07)	(0.04)	
Exposure co-signed ($\times$ contaminated)	0.20	1.69**	0.17**	0.16**	
1TV < 60 HS.RS	(0.22)	(0.30)	(0.03)	(0.04)	
Exposure co-signed	-0.015	-0.067	-0.30**	-0.30**	
1TV < 60 HS.RS	(0.20)	(0.23)	(0.03)	(0.04)	
Exposure co-signed ($\times$ contaminated)	2.24**	1.96**	0.24**	0.24**	
1TV < 60 HS.RS	(0.22)	(0.23)	(0.04)	(0.04)	
Exposure has HELOC	4.56**	4.68**	0.38**	0.38**	
1TV < 60 HS.RS	(0.37)	(0.37)	(0.05)	(0.05)	
Exposure revolving debt utilization > 50%	0.073	0.071	0.037	0.037	
1TV < 60 HS.RS	(0.073)	(0.073)	(0.019)	(0.019)	
1TV < 60 HS.RS	0.891**	0.891**	0.077	0.077	
1TV < 60 HS.RS	(0.10)	(0.10)	(0.024)	(0.024)	
1TV < 95	2.32**	2.31	0.24**	0.24**	
1TV < 95	(0.10)	(0.10)	(0.03)	(0.03)	
Exposure co-signed < 1TV < 60 HS.RS	-0.22	-0.117	0.0038	0.0038	
1TV < 60 HS.RS	(0.21)	(0.21)	(0.06)	(0.06)	
Exposure co-signed < 1TV < 95	1.04	1.12*	0.19**	0.19**	
1TV < 95	(0.40)	(0.40)	(0.03)	(0.03)	
Exposure co-signed < 1TV < 95	1.68**	1.87**	-0.17*	-0.16*	
1TV < 95	(0.37)	(0.37)	(0.07)	(0.07)	
<i>F</i>	0.018	0.018	0.005	0.005	

Note: Table 5 presents results from linear probability model regressions. There are 574,221 observations. In both panels, the dependent variable equals 1 if at least one mortgage delinquency day is observed within the credit period. The standard errors of mortgage origination. For borrowers without parent co-signing, the average delinquency rate is 3.65 percent for all credit products and 0.24 percent for mortgage products; the standard deviations are 18.75 percent and 4.9 percent, respectively. The variable *Parent paid off HELOC/mortgage* is equal to 1 if the co-signing parent had a HELOC/mortgage and paid it off before the mortgage origination. The variable *Parent co-signed* equals 1 if the co-signing parent co-signed the mortgage. The variable *Parent revolving debt utilization* is 250% in equal to 1 if the co-signing parent had a credit balance over credit limit of at least 500 percent, where credit limit is total limit across all revolving credit accounts. All regressions control for LTV, number of borrowers, rate type (FRM vs VM), lender FSA, year-quarter FSA, and FSA FE. Standard errors clustered at FSA level.

	Dependent variable: $\Delta(\text{debt/longevity})$				
	(1)	(2)	(3)	(4)	(5)
$\lambda(\text{parent co-signed})$	0.88*** (0.16)	0.019 (0.25)	0.029 (0.25)	0.025 (0.22)	0.024 (0.25)
$\lambda(\text{parent co-signed}) \times I(\text{constrained})$		-1.15*** (0.27)	-1.46** (0.62)		
$\lambda(\text{parent co-signed}) \times I(\text{FDS})$			2.94*** (0.59)		
$\lambda(\text{parent co-signed}) \times \lambda$				1.51*** (0.30)	
$I(\lambda \in (0, 0.5])$					0.37 (0.36)
$I(\lambda \in (0.5, 0.75])$					0.80 (0.42)
$I(\lambda \in (0.75, 1])$					1.50 (0.30)
χ^2	0.000	0.000	0.000	0.000	0.000

Note: There are 574,221 observations. Dependent variable = delinquency.
 Δ (delinquency) during the 2 years prior origination. Δ (delinquency) during the 2 years pre-origination) $\times 100$, where each indicator variable equals 1 if at least one borrower is 90+ days late on any credit product within the 2-year window. For borrowers without partner co-signing, the dependent variable has a mean of 0.80. The indicator variable 1 ($FTHS$) equals 1 if the FTHBS ratio exceeds the average FTHBS ratio for all borrowers with partner co-signing.
 Δ (price) measures the covariation between the change in house price and the maximum increase in price affordable with combined income. It is close to 1 if FTHBS level up given partner co-signing and purchase near the most expensive house that they can afford. For other FTBHS, Δ (price) equals 0.
All regressions control for LTV, number of borrowers, rate type (FRM vs VRM), property type (condo vs other), lender's fixed effects, year-quarter fixed effects, and PSA fixed effects. Standard errors are clustered by lender.



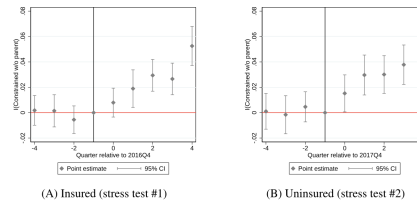


Fig. 11. Event study—Dependent variable: $I(\text{constrained w/o parent co-sign})$.
 Note: The vertical line in panel (A) corresponds to 2016q3, the quarter before the insured stress test was announced (Oct 2016). The vertical line in panel (B) corresponds to 2017q3, the quarter before the uninsured stress test was announced (Oct 2017).

Read:

- Table 3 shows that parent co-signors are more delinquency-prone than non-co-signors, and that the difference is especially strong for constrained borrowers.
- For all credit products, delinquency is about **1.05 percentage points** higher for parent co-signors even before adding the constrained interaction, and the constrained interaction adds another **1.59 percentage points**, for a total effect of about **2.64 percentage points**.
- Table 4 looks at the *change* in delinquency from pre- to post-origination. Parent co-signing is associated with an increase of about **0.88 percentage points**, and the increase is concentrated among constrained borrowers and high- λ borrowers.
- Figure 10 shows that parents with their own mortgage also experience higher delinquency after co-signing with constrained adult children, suggesting that parental exposure is real rather than purely formal.

5.8 Figure 11: stress tests and causal variation in co-signing

Read:

- The 2016 insured and 2018 uninsured stress tests serve as exogenous tightening of PTI constraints.
- In both event studies, the treatment group shows a clear post-policy rise in constrained parent co-signing.
- Four quarters after the policy change, the treatment effect exceeds about **5 percentage points**.
- The paper's interpretation is that tighter underwriting standards cause more marginal borrowers to turn to parents, rather than only reshuffling who gets a mortgage.

6 Short summary

- **Method.** The paper combines a stylized PTI model, rich administrative mortgage and credit data, qualification-rule counterfactuals, and stress-test event studies.
- **Identification.** The housing counterfactual is identified from exact underwriting rules, while the causal interpretation is strengthened by exogenous PTI tightening from the mortgage stress tests.
- **Application.** Canadian FTHBs and parental mortgage co-signing in an era of rapidly rising house prices.

- **Main result.** Parent co-signing substantially relaxes mortgage qualification constraints, but constrained borrowers use that extra qualifying income to lever up and buy much more housing, which raises subsequent financial stress for both children and, in some cases, parents.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Parent co-signing has become an increasingly important way for Canadian FTHBs to overcome PTI constraints.
- This form of parental support is powerful enough to materially change housing choices: for many constrained borrowers, the chosen house would be unattainable without parents.
- The paper's central contribution is to show that co-signing does not merely improve access; it also increases leverage and can raise financial stress after origination.

7.2 Economic intuition

- A binding PTI rule prevents some borrowers from buying the house they would like.
- Adding parent income relaxes the qualification constraint, but the borrower then has a new feasible set and often chooses a **more expensive** house rather than a safer version of the old one.
- Once the mortgage is sized using combined parent-child income, the household may require continuing parental support to keep monthly payments manageable.
- That is why co-signing can simultaneously improve access to homeownership and worsen ex post fragility.

7.3 Takeaway

The paper's main lesson is that **family finance can offset regulatory borrowing constraints, but not necessarily in the way regulators intend**. In this setting, parental co-signing relaxes the mortgage qualification margin, yet the marginal borrower responds by taking on more housing and more leverage. That makes co-signing an important channel of intergenerational support, but also a channel through which housing-market stress is transmitted across generations.